

A More Powerful Selective Inference Test for k -Means Clustering

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Abstract

After partitioning data into clusters with k -means, a natural next step is to test whether two estimated clusters have different means. Because the very hypothesis is chosen by looking at the data, a classical z - or χ^2 -test is badly anti-conservative: on the Palmer penguins data its p -values for three cluster pairs are on the order of 10^{-57} , 10^{-29} , and 10^{-134} . Chen and Witten [2023] solve this by conditioning on the *entire* realized sequence of Lloyd-algorithm cluster assignments and computing an exact truncated χ p -value. Conditioning on the full algorithmic path is convenient but discards power. We make three contributions. First, following the “condition on less” principle of Chen et al. [2023] and the selective optimality theory of Fithian et al. [2017], we propose a test that conditions only on the event that the two *tested* clusters are preserved, taken as a union over all Lloyd paths that produce them; this enlarges the truncation set and is uniformly more powerful, with the gain widening as the number of clusters K grows. Second, when the noise variance is unknown, we develop an exact studentized test built on a Fisher– F pivot, generalizing Yun and Barber [2023] from $K = 2$ to arbitrary K and eliminating their Monte Carlo step; the test requires no estimate of σ at all. Third, we give a fast numerical/analytic solver for the quadratic truncation inequalities that define the conditioning set [Yun and He, 2024]. In simulations the union test controls the selective Type I error (rejection rate 0.035–0.045 at the global null versus 0.975 for the naive test) while substantially improving power, and a head-to-head comparison against data thinning [Neufeld et al., 2024] shows that our studentized test detects real clusters far more often — because data thinning must spend half the data on the test fold and must estimate σ to perform the split, whereas our test needs neither.

1 Introduction

Clustering is routinely used to generate hypotheses: one partitions the observations, notices that two groups look different, and then asks whether the difference is real. The trouble is that the groups, and hence the hypothesis, were chosen using the same data used to test it — a form of double dipping. A classical Wald test that ignores this selection has a wildly inflated Type I error rate [Gao et al., 2024, Chen and Witten, 2023]. The size of the problem is easy to see empirically: clustering the Palmer penguins data into $K = 3$ groups and applying a naive χ^2 test to the three cluster pairs gives p -values of 3.3×10^{-57} , 7.5×10^{-29} , and 5.6×10^{-134} (Section 7); these are absurd certificates of significance produced *purely* by the act of clustering.

Selective inference repairs this by conditioning on the selection event [Lee et al., 2016, Fithian et al., 2017]. For k -means clustering, Chen and Witten [2023] test $H_0 : \boldsymbol{\mu}^\top \boldsymbol{\nu} = \mathbf{0}$ for the contrast $\boldsymbol{\nu}$ encoding the difference in means of two estimated clusters, conditioning on the clustering output (and on nuisance directions) so that the conditional law of the test statistic is a truncated χ distribution. Their test controls the selective Type I error and is computable in closed form.

However, it conditions on the *full realized Lloyd path*: the entire sequence of intermediate cluster assignments produced by the algorithm, iteration by iteration. This is far more information than the hypothesis requires — the hypothesis only concerns the two clusters that were ultimately tested — and conditioning on it shrinks the truncation set and the power.

Contributions. This paper makes three contributions.

1. **A more powerful test that conditions on less** (Section 3). We condition only on the event that the two *tested* clusters are preserved, unioned over every Lloyd path that outputs them. This is the k -means analogue of the “condition on less $\Rightarrow$ more power” principle that Chen et al. [2023] used for the graph fused lasso, and is justified by the selective-optimality framework of Fithian et al. [2017]. The resulting truncation set is a superset of the path set, so the test is uniformly more powerful; empirically the power gain over the path test widens with the number of clusters K .
2. **An exact studentized test for unknown variance** (Section 4). Following Yun and Barber [2023] we studentize, obtaining an F pivot in which σ cancels. We characterize the conditioning set *exactly* on the “ R -fiber” — a one-parameter rotation of the data at fixed total energy — for any K , replacing the importance sampling that Yun and Barber [2023] require for $K > 2$ with an exact, zero-variance p -value.
3. **A fast solver** (Section 5). The conditioning set is the solution set of a collection of quadratic inequalities in the perturbation parameter; we compute the coefficients analytically and solve by an interval/scan method, in the spirit of Yun and He [2024].

We also report a head-to-head comparison against data thinning [Neufeld et al., 2024, Dharamshi et al., 2024], a sample-splitting alternative to selective inference. Data thinning is valid, but it pays two prices that our studentized test avoids: it clusters on only half the data, so it detects fewer real clusters, and it needs an estimate of σ to perform the split — an estimate that is inflated two- to three-fold under genuine cluster separation, poisoning the split.

2 Background: the path test of Chen and Witten

2.1 Model and hypothesis

We observe $\mathbf{X} \in \mathbb{R}^{n \times q}$ with independent rows,

$$\mathbf{X} \sim \mathcal{MN}_{n \times q}(\boldsymbol{\mu}, \mathbf{I}_n, \sigma^2 \mathbf{I}_q), \quad \text{i.e.} \quad X_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \sigma^2 \mathbf{I}_q), \quad (1)$$

with unknown mean rows $\boldsymbol{\mu}_i$ and (for now) known σ^2 . Running k -means with K clusters returns a partition $\hat{\mathcal{C}}_1, \dots, \hat{\mathcal{C}}_K$ of $\{1, \dots, n\}$. Given two estimated clusters $\hat{\mathcal{C}}_1, \hat{\mathcal{C}}_2$ we test

$$H_0 : \frac{1}{|\hat{\mathcal{C}}_1|} \sum_{i \in \hat{\mathcal{C}}_1} \boldsymbol{\mu}_i = \frac{1}{|\hat{\mathcal{C}}_2|} \sum_{i \in \hat{\mathcal{C}}_2} \boldsymbol{\mu}_i \iff H_0 : \boldsymbol{\mu}^\top \boldsymbol{\nu} = \mathbf{0}_q, \quad (2)$$

where the contrast $\boldsymbol{\nu} \in \mathbb{R}^n$ has entries $\nu_i = \mathbf{1}\{i \in \hat{\mathcal{C}}_1\}/|\hat{\mathcal{C}}_1| - \mathbf{1}\{i \in \hat{\mathcal{C}}_2\}/|\hat{\mathcal{C}}_2|$. The natural statistic is $\|\mathbf{X}^\top \boldsymbol{\nu}\|_2$, the length of the estimated difference in means.

2.2 The selective p -value and the Lloyd path

The naive Wald p -value $p_{\text{naive}} = \mathbb{P}(\|\mathbf{X}^\top \boldsymbol{\nu}\|_2 \geq \|\mathbf{x}^\top \boldsymbol{\nu}\|_2)$ ignores that $\boldsymbol{\nu}$ was chosen by clustering and is severely anti-conservative. Chen and Witten [2023] instead condition on the clustering output. Writing $c_i^{(t)}(\mathbf{X})$ for the cluster assigned to observation i at iteration t of Lloyd’s algorithm, they define

$$p_{\text{path}} = \mathbb{P}_{H_0} \left(\left\| \mathbf{X}^\top \boldsymbol{\nu} \right\|_2 \geq \left\| \mathbf{x}^\top \boldsymbol{\nu} \right\|_2 \mid \bigcap_{t=0}^T \bigcap_{i=1}^n \{c_i^{(t)}(\mathbf{X}) = c_i^{(t)}(\mathbf{x})\}, \mathbf{P}_\nu^\perp \mathbf{X} = \mathbf{P}_\nu^\perp \mathbf{x}, \text{dir}(\mathbf{X}^\top \boldsymbol{\nu}) = \text{dir}(\mathbf{x}^\top \boldsymbol{\nu}) \right). \quad (3)$$

The conditioning event fixes everything except a single scalar: perturbing the data along $\boldsymbol{\nu} \text{dir}(\mathbf{x}^\top \boldsymbol{\nu})^\top$, $\mathbf{x}'(\phi) = \mathbf{x} + ((\phi - \|\mathbf{x}^\top \boldsymbol{\nu}\|_2) / \|\boldsymbol{\nu}\|_2^2) \boldsymbol{\nu} \text{dir}(\mathbf{x}^\top \boldsymbol{\nu})^\top$, the p -value reduces to a one-dimensional truncated- χ probability,

$$p_{\text{path}} = \mathbb{P}(\phi \geq \|\mathbf{x}^\top \boldsymbol{\nu}\|_2 \mid \phi \in \mathcal{S}_T), \quad \phi \sim (\sigma \|\boldsymbol{\nu}\|_2) \cdot \chi_q, \quad (4)$$

where $\mathcal{S}_T = \{\phi \geq 0 : \text{the entire Lloyd path is preserved at } \mathbf{x}'(\phi)\}$. The key computational fact is that $\mathcal{S}_T$ is the intersection of the solution sets of $O(nKT)$ quadratic inequalities in ϕ , so it is a union of intervals computable analytically; the test controls the selective Type I error in the sense of Lee et al. [2016] and Fithian et al. [2017].

2.3 The orthogonal decomposition

A geometric view, which we use throughout, decomposes the data with three Frobenius-orthogonal projections in $\mathbb{R}^{n \times n}$,

$$\mathbf{X} = \mathbf{P}_0 \mathbf{X} + \mathbf{P}_1 \mathbf{X} + \mathbf{P}_2 \mathbf{X}, \quad (5)$$

where (i) the rank-one *between-cluster* projection $\mathbf{P}_0 = \boldsymbol{\nu} \boldsymbol{\nu}^\top / \|\boldsymbol{\nu}\|_2^2$ carries the contrast, with $\|\mathbf{P}_0 \mathbf{X}\|_F = \|\mathbf{X}^\top \boldsymbol{\nu}\|_2 / \|\boldsymbol{\nu}\|_2$; (ii) the *within-cluster* centering projection $\mathbf{P}_1$ of the two tested clusters (rank $m - 2$, with $m = |\hat{\mathcal{C}}_1| + |\hat{\mathcal{C}}_2|$) carries the within-cluster scatter; and (iii) the *nuisance* $\mathbf{P}_2 = \mathbf{I} - \mathbf{P}_0 - \mathbf{P}_1$. The path test moves only $\|\mathbf{P}_0 \mathbf{X}\|_F$ along a ray, holding $\mathbf{P}_1 \mathbf{X}$ and $\mathbf{P}_2 \mathbf{X}$ fixed; Figure 2 contrasts this with the studentized test of Section 4.

Why the path test loses power. Conditioning on the whole path is *sufficient* for validity but not *necessary*: the hypothesis depends only on the two tested clusters, not on the order in which Lloyd’s algorithm reached them. Many distinct intermediate assignment sequences end in the same final partition of the two tested clusters; the path test discards all of them but the realized one. By Fithian et al. [2017], conditioning on this extra, ancillary information can only cost power.

3 A more powerful test: condition on less

3.1 The union conditioning set

Our test conditions on the *final-partition* event for the two tested clusters, rather than the full path. Define

$$\mathcal{S}_{\text{union}} = \{\phi \geq 0 : k\text{-means on } \mathbf{x}'(\phi) \text{ outputs clusters that preserve both } \hat{\mathcal{C}}_1 \text{ and } \hat{\mathcal{C}}_2\}. \quad (6)$$

Concretely, along the perturbation $\mathbf{x}'(\phi)$ we re-run k -means (with a shared initialization) and record, for each ϕ , whether the two tested clusters reappear — regardless of which Lloyd path produced them. Because the realized path is one of the paths that preserve the tested clusters,

$$\mathcal{S}_T \subseteq \mathcal{S}_{\text{union}}, \quad (7)$$

so $\mathcal{S}_{\text{union}}$ is a superset of the path set, and the corresponding truncated- χ p -value, $p_{\text{union}} = \mathbb{P}(\phi \geq \|\mathbf{x}^\top \boldsymbol{\nu}\|_2 \mid \phi \in \mathcal{S}_{\text{union}})$, is based on weaker conditioning. This is exactly the construction Chen et al. [2023] use for the graph fused lasso, transported to k -means: condition on the *tested object* (here, the two clusters) instead of the *algorithmic trace* (the Lloyd path).

3.2 Validity

The union set is, like the path set, a function of ϕ alone along the perturbation line (everything orthogonal to the contrast is held fixed), so $\phi \mid \phi \in \mathcal{S}_{\text{union}}$ is again a truncated scaled- χ_q and the test controls the selective Type I error. Empirically, at the global null the union test rejects at rate 0.045 (KS-uniform), versus 0.975 for the naive test (Table 1; `power_type1_summary.rds`).

3.3 Construction of the union set

We enumerate the distinct Lloyd paths reachable along the perturbation line as follows. Starting from the observed $\phi = \|\mathbf{x}^\top \boldsymbol{\nu}\|_2$, we compute the ϕ -interval on which the observed path is realized (the path test’s $\mathcal{S}_T$), then probe just past each interval endpoint, re-run k -means there to discover the neighbouring path, compute *its* preservation interval, and repeat, queueing newly discovered endpoints. The union of all intervals whose path preserves both tested clusters is $\mathcal{S}_{\text{union}}$; the subset of intervals matching the observed path signature recovers $\mathcal{S}_T$ exactly, so the two p -values are produced from a single sweep. The exploration is purely Euclidean (it perturbs $\mathbf{x}$ and re-runs Euclidean k -means), which makes it *metric-independent*: the known general-covariance test $\mathbf{X} \sim \mathcal{MN}(\boldsymbol{\mu}, \mathbf{I}_n, \boldsymbol{\Sigma})$ reuses the identical exploration and only rescales the resulting intervals by a scalar and swaps the test statistic for $\sqrt{(\mathbf{x}^\top \boldsymbol{\nu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}^\top \boldsymbol{\nu})}$. The general-covariance path p -value matches the isotropic one to machine precision.

A scan cutoff. The far tail of the perturbation line contributes negligible χ mass to the p -value. We therefore stop exploring once ϕ exceeds the point where the conditional χ survival mass beyond it falls below a fraction `scan_tail_prob` (default 10^{-6}) of the survival mass beyond the observed statistic. This leaves the p -value essentially unchanged (verified $\Delta \leq 10^{-11}$) while cutting roughly 25–40% of the path exploration.

4 Unknown variance: the studentized R -fiber

4.1 The F pivot

When σ is unknown we cannot compare $\|\mathbf{x}^\top \boldsymbol{\nu}\|_2$ to a scaled χ_q , because the scale is exactly what we lack. Following Yun and Barber [2023] we studentize. Using the decomposition of Section 2.3, the within-cluster scatter $\|\mathbf{P}_1 \mathbf{X}\|_F^2$ is an unbiased σ^2 -estimate independent of the between-cluster signal $\|\mathbf{P}_0 \mathbf{X}\|_F^2$, so their ratio is pivotal:

$$R = \frac{\|\mathbf{P}_0 \mathbf{X}\|_F^2 / q}{\|\mathbf{P}_1 \mathbf{X}\|_F^2 / ((m-2)q)} = (m-2) \frac{\|\mathbf{P}_0 \mathbf{X}\|_F^2}{\|\mathbf{P}_1 \mathbf{X}\|_F^2} \sim F_{q, (m-2)q} \quad \text{under } H'_0, \quad (8)$$

where $m = |\hat{\mathcal{C}}_1| + |\hat{\mathcal{C}}_2|$ counts only the two tested clusters and the unknown σ cancels. The required null is the slightly stronger *pointwise* null H'_0 : within each tested cluster the means $\boldsymbol{\mu}_i$ are equal to the cluster mean (so that $\|\mathbf{P}_1 \mathbf{X}\|_F^2$ is a clean noise estimate). This is the price of not knowing σ .

4.2 The rotation and the fiber

We condition on the clustering, on $\mathbf{P}_2\mathbf{X}$, on the unit directions $U_0 = \mathbf{P}_0\mathbf{X}/\|\mathbf{P}_0\mathbf{X}\|_F$ and $U_1 = \mathbf{P}_1\mathbf{X}/\|\mathbf{P}_1\mathbf{X}\|_F$ (which are orthonormal), and on the *total energy* $T = \|\mathbf{P}_0\mathbf{X}\|_F^2 + \|\mathbf{P}_1\mathbf{X}\|_F^2$. The single remaining degree of freedom is an angle θ , and the conditional perturbation is

$$\mathbf{x}(\theta) = \mathbf{P}_2\mathbf{x} + \sqrt{T}(\sin\theta U_0 + \cos\theta U_1), \quad \mathbf{x}(\theta_{\text{obs}}) = \mathbf{x}, \quad (9)$$

with $\theta_{\text{obs}} = \text{atan2}(\|\mathbf{P}_0\mathbf{x}\|_F, \|\mathbf{P}_1\mathbf{x}\|_F)$ and $R(\theta) = (m-2)\tan^2\theta$. Because $U_0 \perp U_1$ are orthonormal, (9) is a **Givens rotation** of the vectorized data $\mathbf{x} \in \mathbb{R}^{nq}$ inside the plane $\text{span}\{U_0, U_1\}$, fixing its orthogonal complement,

$$\mathbf{x}(\theta) = G(\theta - \theta_{\text{obs}})\mathbf{x}, \quad G(\delta) = \mathbf{I} + (\cos\delta - 1)(U_0U_0^\top + U_1U_1^\top) + \sin\delta(U_0U_1^\top - U_1U_0^\top). \quad (10)$$

Since G is orthogonal it preserves the in-plane radius, so T is held fixed and only the ratio R varies — which is precisely why the test is free of σ : T is complete and sufficient for σ under H'_0 (Basu's theorem), so the scale-invariant ratio R is ancillary to it. Contrast this with the known- σ ray $\mathbf{x}'(\phi)$, which moves $\|\mathbf{P}_0\mathbf{X}\|_F$ in a straight line while $\mathbf{P}_1\mathbf{X}$ stays put. The two slices are drawn in Figure 2: the known- σ test walks a horizontal ray (fix within-scatter, vary separation $\rightarrow \chi$), the studentized test walks an arc (fix total energy, vary the angle $\rightarrow F$).

Geometry. Datasets of fixed energy T lie on a sphere of radius $\sqrt{T}$. The pole ($\theta = \pi/2$) is all separation (within-scatter 0), the equator ($\theta = 0$) is fully merged clusters (no separation), the latitude is θ (and $R = (m-2)\tan^2\theta$ depends only on latitude), and the longitude indexes the within-scatter pattern. Our test fixes the longitude and sweeps θ : it walks a single meridian, the latitude playing the role of the test statistic.

4.3 The truncated- F test and the union

With $\mathcal{S} = \{\theta \in (0, \pi/2) : \text{the clustering is preserved at } \mathbf{x}(\theta)\}$, the selective p -value is the truncated- F probability

$$p = \mathbb{P}(R(\theta) \geq R \mid \theta \in \mathcal{S}), \quad R \sim F_{q, (m-2)q}. \quad (11)$$

The cluster-assignment margins $\|x_i - m_j\|^2 - \|x_i - m_k\|^2$ along the fiber are quadratic forms in $(\sin\theta, \cos\theta)$, so $\mathcal{S}$ is a union of arcs computed exactly (Section 5). Unioning the arcs across all Lloyd paths that preserve the two tested clusters yields the more powerful union version — and, crucially, this union is computed *on the same R -fiber*, so the truncated- F remains a valid pivot over it.

An important caveat: do not remap the known- σ union. A natural shortcut is to take the known- σ ϕ -union set and feed it, re-scaled, into the F pivot. This is *invalid*: the studentized- F pivot holds only *per Lloyd path*, and the per-path truncated- F 's do not aggregate into a union- F unless the union is taken in the F pivot's own coordinate. Along the ϕ -ray the within-scatter $\|\mathbf{P}_1\mathbf{X}\|_F$ is held fixed, but along the R -fiber it varies, so the per-path cancellation that makes R pivotal breaks when one imports the ϕ -fiber's set. Exhaustive simulation confirms the remapped union over-rejects (rejection 0.081, scaling with union width from 0.010 up to 0.170), whereas the same union taken on the R -fiber is calibrated. The valid construction perturbs along the R -fiber (9) and unions the θ -arcs, so that $\{\text{clusters preserved}\} = \{R \in \mathcal{S}'\}$ is a function of R alone and the union of a one-dimensional truncation is exact.

Generalizing Yun and Barber to any K . For $K = 2$ the two tested clusters are all the data ($m = n$) and Yun and Barber [2023] compute the truncation set exactly by reusing Chen and Witten [2023]. For $K > 2$ they resort to importance sampling. Our exact R -fiber solver computes the arc set exactly for any K , so it (i) replaces their Monte Carlo with an exact, zero-variance p -value for $K > 2$ k -means (even for the path test), and (ii) yields the more powerful union for free. The method is k -means-specific (theirs applies to general clustering, e.g. hierarchical) and requires the pointwise null H'_0 .

When does the union help under unknown variance? The union’s power gain is largely a *known-variance* phenomenon. A one-sided truncated p -value shrinks when enlarging the truncation set only if the added mass lands *below* the observed statistic. For the known- σ χ statistic, alternative Lloyd paths tolerate smaller separation and so extend the ϕ -interval downward, below the observed value — producing large gains. For the studentized F , k -means has *minimized* the within-scatter $\|\mathbf{P}_1 \mathbf{X}\|_F$, pinning R_{obs} against its low- R floor, so the extra room is almost entirely *above* the observed statistic and the union p -value $\approx$ the path p -value. In short, the union’s leverage is *scale* (contrast-length) information, which a scale-free statistic discards. This is why we report the union gain primarily under known σ (Figure 3) and report the studentized test primarily as the σ -free route to validity and to the Yun–Barber power regime (Figure 4).

5 Computation

Both conditioning sets reduce to solving quadratic inequalities in a single perturbation parameter.

Known σ (ϕ -ray). Along $\mathbf{x}'(\phi)$ each k -means assignment margin is a quadratic in ϕ , so the per-path preservation set is the intersection of $O(nKT)$ quadratic inequalities, computed in closed form, and the union set is assembled by the endpoint-probing sweep of Section 3.

Unknown σ (R -fiber). Along $\mathbf{x}(\theta)$ each assignment margin is an exact quadratic form $Q(a, b) = Aa^2 + Bb^2 + Cab + Da + Eb + F_c$ in $(a, b) = (\sin \theta, \cos \theta)$. Substituting $t = \tan(\theta/2)$ turns $Q(\sin \theta, \cos \theta)(1 + t^2)^2$ into a *quartic* $N(t)$, whose real roots in $[0, 1]$ give the arc breakpoints; the sign of Q at a midpoint decides which arcs satisfy the inequality, and intersecting over all margins yields the arc set for one Lloyd trajectory (`sims/rfiber_exact.R`). The truncated- F mass over the resulting r -intervals is computed in a cancellation-proof way (taking the maximum of the CDF-difference and the survival-difference, the latter stable in the extreme upper tail under strong separation). The solver is modular and unit-tested: 23 tests for the F -region routine (including agreement with Monte Carlo and extreme-tail underflow), and the arc assembler passes its trajectory-preservation unit tests; the truncated- F pivot is verified uniform under fixed truncation. A pure interval/scan implementation also computes the set without analytic quartics, agreeing with the exact solver where the grid is fine and finding more arcs where the grid is coarse; numerically these two solvers agree. The same numerical-solver idea underlies the multiple-pairs extension of Yun and He [2024].

6 Simulations

Unless noted, data are q -dimensional Gaussian with isotropic unit variance; k -means uses a fixed shared initialization. We report rejection rates at level 0.05.

Table 1: Selective Type I error at the global null ($\mu = \mathbf{0}$). Rejection rates at level 0.05. The naive test is grossly invalid; the selective tests are calibrated. Known- σ union/path from `power_type1_summary.rds` ($n = 20K$, $q = 2$, 200 reps); studentized (R -fiber) and data-thinning columns from the null cell of `datathin_compare.rds` ($K = 3$, $q = 2$, 400 reps).

	naive	path (known)	union (known)	studentized F	data thinning (oracle)
rejection rate	0.975	0.045–0.060	0.035–0.045	0.040	0.065

Table 2: Power of the known- σ union vs. path test against separation δ ($n = 20K$, $q = 2$, $K = 3$; `power_type1_summary.rds`, 200 reps). The union is uniformly more powerful. Figure 3.

δ	0	1	2	3	4	5	6
union	0.045	0.075	0.195	0.445	0.660	0.910	0.925
path	0.060	0.060	0.085	0.230	0.415	0.645	0.770

6.1 Type I error

At the global null ($\mu = \mathbf{0}$) the union, path, and studentized tests are all calibrated, while the naive test is invalid. From the global-null cell of `power_type1_summary.rds` (200 replicates), the union rejects at 0.045 and the path at 0.060; the naive test rejects at 0.975 (Table 1). The studentized R -fiber test is likewise calibrated across $q \in \{2, 10, 50, 100\}$ (Type I 0.047/0.046/0.029/0.054; slightly conservative at high q). Under AR(0.5)/AR(0.9) covariance the general-covariance union and path remain on the diagonal while the naive test flatlines (Figure 5). Under heavy-tailed t_5/t_{10} noise the studentized test is fairly robust, with mild anti-conservatism only for the heaviest tails (Figure 6).

6.2 Power: union versus path (known σ)

Table 2 reports power for the known- σ union and path tests as a function of the separation δ (from `power_type1_summary.rds`). The union is uniformly more powerful, with gains of order 0.1–0.3; averaged over $\delta > 0$ the union improves on the path by about 16.8 percentage points. The gain *widens with* K : at $\delta = 5$, $q = 2$ the union’s advantage over the path is 0.075/0.115/0.225/0.200 for $K = 2/3/4/5$. This is a distinctive feature of our exact solver, which handles any K .

6.3 Unknown variance: reproducing the Yun–Barber regime

We reproduce the equilateral-triangle, $K = 3$, $q \in \{2, 10\}$ design of Yun and Barber [2023] in our k -means framework (path test, clusters 1 vs 2, 300 reps; `repro_yunbarber_q{2,10}.rds`), comparing four variance handlings on the *same* truncation set: an oracle (known σ), the studentized F , a median-based $\hat{\sigma}_{\text{MED}}$ plug-in, and a sample-SD plug-in. The ordering oracle $\geq$ studentized $\geq$ plug-ins holds throughout. The studentized F decisively beats the sample plug-in once the clusters separate: at $q = 2$, $\delta = 7$, power is 0.490 (studentized) versus 0.290 (sample) and 0.717 (oracle); the gap opens at $\delta \approx 4$ –5. The mechanism is that the global sample SD absorbs the between-cluster separation, so $\mathbb{E}[\hat{\sigma}_{\text{sample}}]$ climbs roughly linearly in δ and the plug-in power plateaus, whereas the within-cluster scale stays near 1 and the F approaches the oracle. The robust $\hat{\sigma}_{\text{MED}}$ is dimension-dependent: near-oracle at $q = 10$ but, with three balanced clusters, as inflated as the sample SD at $q = 2$.

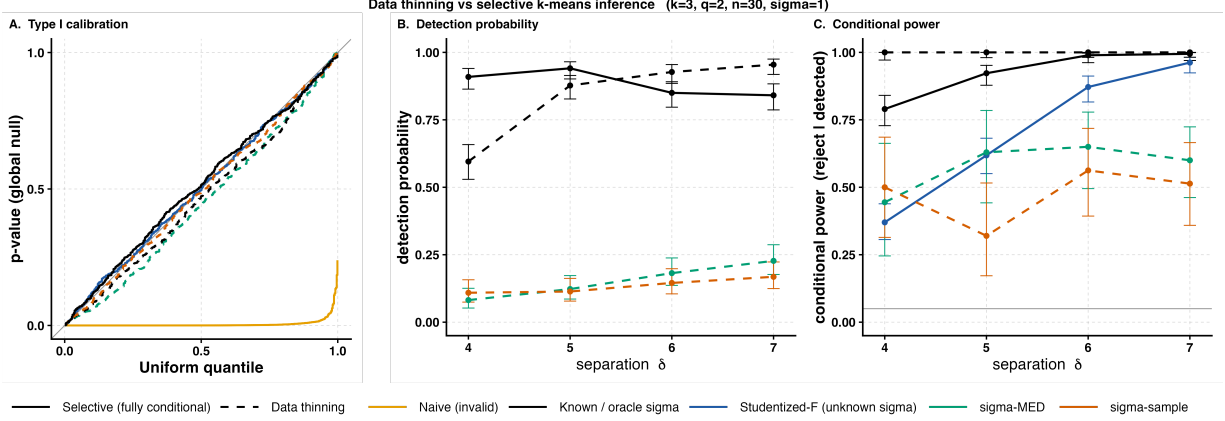


Figure 1: Data thinning vs. the selective tests, all three metrics in the paper’s shared grammar (colour = variance handling; line style = paradigm, selective solid vs. data thinning dashed). **A:** Type I calibration as a QQ plot against the uniform — every valid method, selective *and* thinning, tracks the 45° line, while the naive test (orange) plunges below it. **B:** detection probability vs. separation δ — full-data selective clustering (solid black) dominates the thinned splits, and the plug-in splits ($\hat{\sigma}_{\text{MED}}$ green, $\hat{\sigma}_{\text{sample}}$ vermilion) collapse because the variance estimate is inflated by the very separation being tested. **C:** conditional power vs. δ (reject | detected); oracle thinning rejects whenever it detects, but rarely detects. Cf. Table 3. Source: `sims/plot_datathin.R` → `datathin.png`.

6.4 Comparison with data thinning

Data thinning [Neufeld et al., 2024] is a sample-splitting alternative: split each entry $X = X^{(1)} + X^{(2)}$ into independent Gaussian halves, cluster on $X^{(1)}$, and test on $X^{(2)}$ with an ordinary (unconditional) mean test. It is valid by construction, but it pays two prices our studentized test avoids. *First*, it clusters on only half the information, so it recovers the true pair less often. *Second*, to perform the split it needs a variance: the oracle (σ^2) is unavailable in practice, and any plug-in is inflated under cluster separation, which poisons both folds.

Table 3 reports the head-to-head, per-replicate matched, on a $K = 3$ equilateral-triangle design ($q = 2$, $n_{\text{per}} = 10$, $\varepsilon = 0.5$; `datathin.compare.rds`, null 400 reps, signal 220 reps each). Three things stand out. (i) At the global null all selective tests and oracle thinning control Type I; the plug-in thinning variants are slightly anti-conservative (0.075 for the MED split). (ii) *Detection probability* — whether the clustering recovers the true pair — is far higher for the full-data selective pipeline (e.g. 0.91 at $\delta = 4$) than for plug-in thinning, which clusters on half the data *and* with an inflated split variance, collapsing detection to 0.08–0.23. (iii) Even *conditional* on detection, the selective union and studentized tests are competitive with or better than plug-in thinning, while needing no σ . The recorded variance estimates make the poisoning explicit: under separation the MED and sample estimates inflate to 2.08–3.12 (true $\sigma = 1$) as δ runs $4 \rightarrow 7$. Only oracle thinning — which knows the true σ and so is not realizable — keeps high detection. Our studentized test needs no σ at all and clusters on the full data. Figure 1 draws all three metrics in the paper’s shared grammar (colour = variance handling; line style = paradigm, selective solid vs. thinning dashed): panel A shows the Type I calibration as a QQ plot against the uniform, where every valid method — selective *and* thinning — tracks the 45° line while the naive test plunges below it, and panels B–C plot detection probability and conditional power against δ .

Table 3: Data thinning vs. the selective tests, matched per replicate (`datathin_compare.rds`; $K = 3$, $q = 2$, $n_{\text{per}} = 10$, $\varepsilon = 0.5$). “detect” is the probability the true pair is recovered; “cpow” is rejection rate *conditional* on detection. Oracle thinning uses the true σ (not realizable); MED/sample thinning estimate σ to split. The median split-variance estimates ($\hat{\sigma}_{\text{MED}}$, true $\sigma = 1$) show the inflation that poisons the folds.

δ	4	5	6	7
$\hat{\sigma}_{\text{MED}}$ (true = 1)	2.08	2.42	2.77	3.12
detect selective (full data)	0.91	0.94	0.85	0.84
detect thinning, oracle σ	0.60	0.88	0.93	0.95
detect thinning, MED split	0.08	0.12	0.18	0.23
detect thinning, sample split	0.11	0.11	0.15	0.17
cpow union (known σ)	0.79	0.92	0.99	0.99
cpow studentized F (no σ)	0.37	0.62	0.87	0.96
cpow thinning, oracle σ	1.00	1.00	1.00	1.00
cpow thinning, MED split	0.44	0.63	0.65	0.60
cpow thinning, sample split	0.50	0.32	0.56	0.51

Table 4: Palmer penguins, $K = 3$, p -values for the three cluster pairs (`penguins_pvalues_k3.csv`). The naive test is absurd; the path test rejects none; the union recovers pair 2 vs 3; the studentized test rejects all three. Figure 7.

pair	naive	path (known)	union (known)	studentized F
1 vs 2	3.3×10^{-57}	0.920	0.063	2.3×10^{-5}
1 vs 3	7.5×10^{-29}	0.514	0.271	0.028
2 vs 3	5.6×10^{-134}	0.808	1.6×10^{-14}	1.4×10^{-5}

7 Real data

7.1 Palmer penguins

We cluster the Palmer penguins data [Horst et al., 2020] — four standardized morphological features, the same data used by Yun and Barber [2023] — into $K = 3$ groups and test all three cluster pairs (`penguins_pvalues_k3.csv`, $\hat{\sigma}_{\text{MED}} = 1.155$). Table 4 shows the pattern. The naive test reports absurd p -values (3.3×10^{-57} , 7.5×10^{-29} , 5.6×10^{-134}). The original known- σ *path* test, with the MED plug-in, fails to reject *any* pair (0.92, 0.51, 0.81) — conditioning too restrictively, and with an inflated $\hat{\sigma}$. The known- σ *union* recovers the clearest split (2 vs 3: $0.81 \rightarrow 1.6 \times 10^{-14}$) while staying cautious on the closer pairs (0.063, 0.27). The studentized test, needing no σ , rejects all three (2.3×10^{-5} , 0.028, 1.4×10^{-5}).

7.2 Single-cell RNA sequencing (Pollen)

We cluster 216 developing-cortex cells from Pollen et al. [2014] (balanced across four inferred cell types; CP10k normalization, log1p, top-500 highly variable genes, ten standardized PCs) into $K = 3$ groups and test the three pairs (`scrna_pvalues_k3.csv`). Here the union is the hero: the original *path* test rejects *none* of the three pairs (0.475, 0.179, 0.067), whereas the *union* rejects *all three*

Table 5: Pollen scRNA-seq, $K = 3$, p -values for the three cluster pairs (`scrna_pvalues_k3.csv`). The path test rejects none; the union rejects all three. Figure 8.

pair	naive	path (known)	union (known)	studentized F
1 vs 2	4.6×10^{-34}	0.475	0.0061	0.024
1 vs 3	3.5×10^{-31}	0.179	0.041	1.000
2 vs 3	1.6×10^{-27}	0.067	0.043	1.000

(0.0061, 0.041, 0.043). The studentized test rejects the clearest pair (1 vs 2: 0.024), and the naive p -values are again absurd (10^{-34} , 10^{-31} , 10^{-27}). Penguins and scRNA are complementary: on penguins the studentized test rejects the most pairs, on scRNA the known- σ union does — but in both cases the original path test is too conservative to be useful.

8 Discussion

We have given a more powerful selective test for k -means clustering that conditions on the two tested clusters rather than the full Lloyd path, an exact studentized version that needs no estimate of σ and extends Yun and Barber [2023] to any K , and a fast solver for both. The σ -free property is the practical headline: the most common failure of plug-in approaches — including data thinning’s split step — is that a global variance estimate is inflated by exactly the cluster separation one is trying to detect, and the studentized test sidesteps this entirely.

Several limitations remain. The studentized test requires the pointwise null H'_0 (equal means *within* each tested cluster), stronger than the equal- cluster-means null; this is the inherent cost of not knowing σ , and a violation is conservative unless within-cluster heterogeneity deflates the residual under a true mean difference. The union’s power gain is a known- σ phenomenon and is “studentized away” under the F pivot, because k -means minimizes the within-scatter and pins the statistic against its floor. The method is specific to k -means / Lloyd’s algorithm, whereas Yun and Barber [2023] and Gao et al. [2024] cover broader clustering families.

Figures

References

- Yiqun T. Chen and Daniela M. Witten. Selective inference for k -means clustering. *Journal of Machine Learning Research*, 24(152):1–41, 2023. URL <https://jmlr.org/papers/v24/22-0371.html>. arXiv:2203.15267.
- Yiqun T. Chen, Sean W. Jewell, and Daniela M. Witten. More powerful selective inference for the graph fused lasso. *Journal of Computational and Graphical Statistics*, 32(2):577–587, 2023. doi: 10.1080/10618600.2022.2097246. arXiv:2109.10451.
- Ameer Dharamshi, Anna Neufeld, Keshav Motwani, Lucy L. Gao, Daniela Witten, and Jacob Bien. Generalized data thinning using sufficient statistics. arXiv preprint, 2024. URL <https://arxiv.org/abs/2303.12931>. arXiv:2303.12931.
- William Fithian, Dennis Sun, and Jonathan Taylor. Optimal inference after model selection. *arXiv preprint arXiv:1410.2597*, 2017. First version 2014; revised 2017.

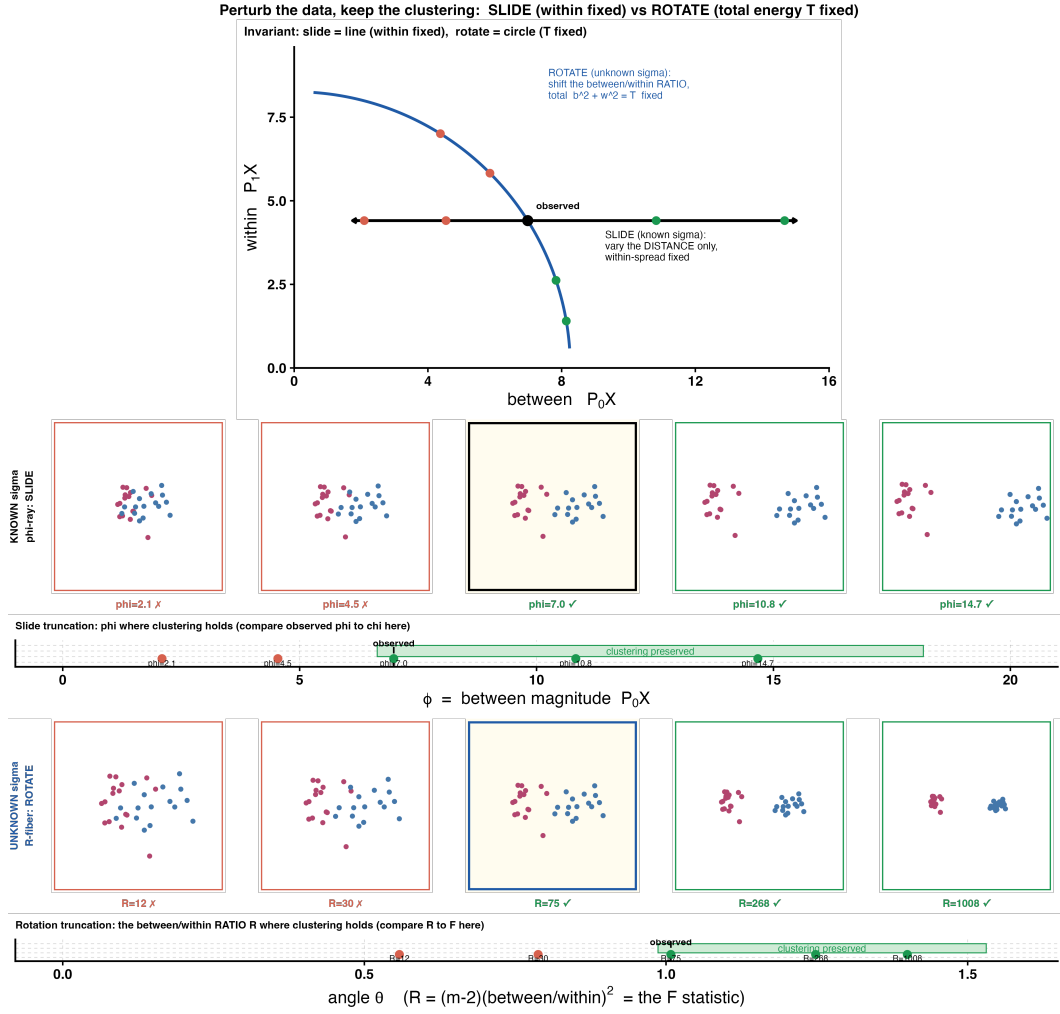


Figure 2: The contrast plane $(\|\mathbf{P}_0\mathbf{X}\|_F, \|\mathbf{P}_1\mathbf{X}\|_F)$. The known- σ ϕ -ray (fix within-scatter, vary separation $\rightarrow \chi$) versus the studentized R -fiber arc (fix total energy T , vary the angle $\rightarrow F$). Source: `sims/plot_decomposition.R` $\rightarrow$ `decomposition.png`.

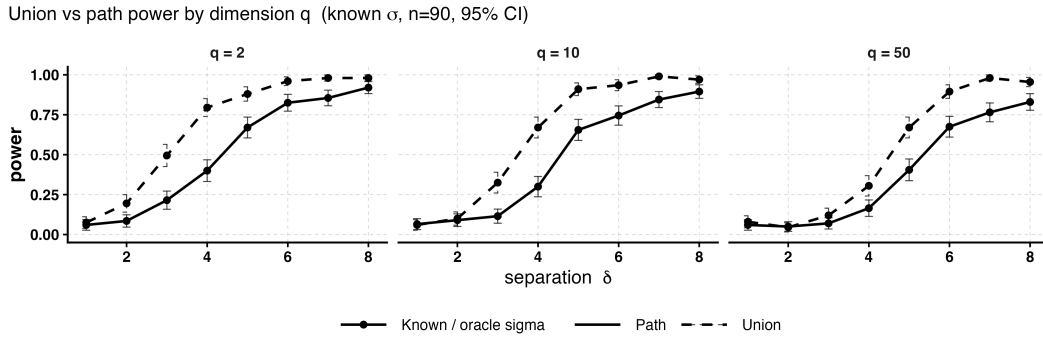


Figure 3: Known- σ power: union (dashed) vs. path (solid) against separation, faceted by q . The union is uniformly more powerful (Table 2); the gain widens with K . Source: `cw_power.png`.

Cost of unknown variance -- same conditioning set, vary only sigma-handling

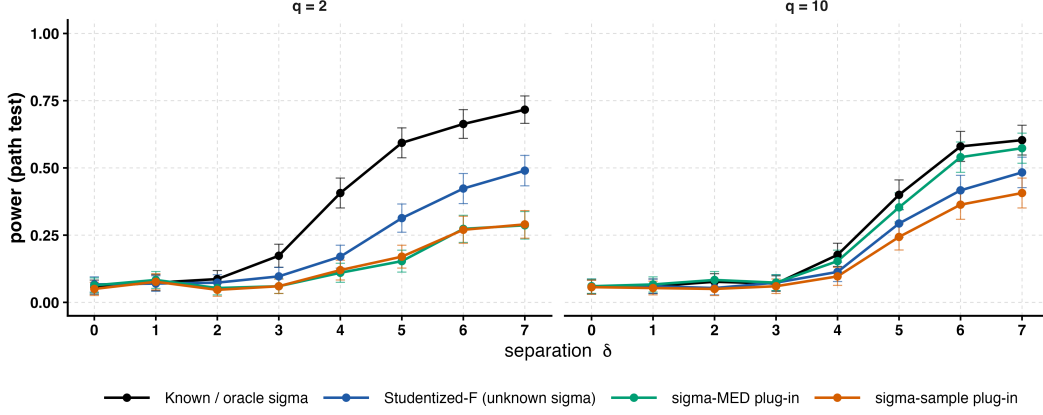


Figure 4: Reproducing the Yun–Barber regime on the same truncation set: oracle $\geq$ studentized $F \geq$ plug-ins. The studentized F beats the sample plug-in once clusters separate, because the global SD absorbs the separation. Source: `repro_yunbarber.png`.

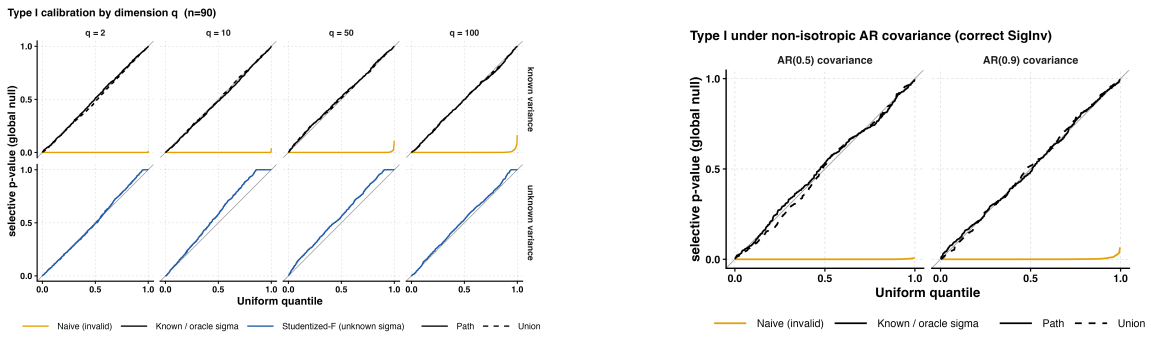


Figure 5: Type I control. Left: QQ plot, selective tests on the diagonal, naive invalid (`cw_typeI.png`). Right: under AR(0.5)/AR(0.9) covariance, the general-covariance union/path stay calibrated (`gencov_typeI.png`).

Type I under heavy-tailed noise (t_5, t_{10}): selective tests calibrated, naive invalid

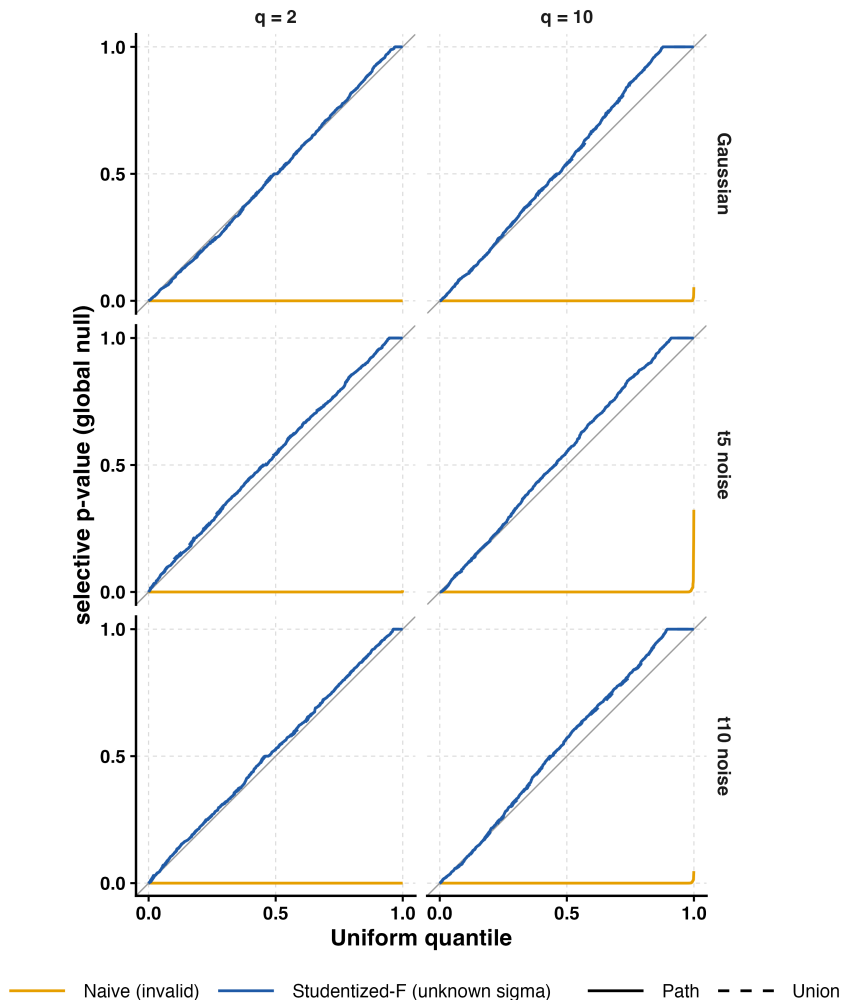


Figure 6: Type I under heavy-tailed t_5/t_{10} noise. The studentized test is fairly robust; mild anti-conservatism only for the heaviest tails. Source: `heavytail_typeI.png`.

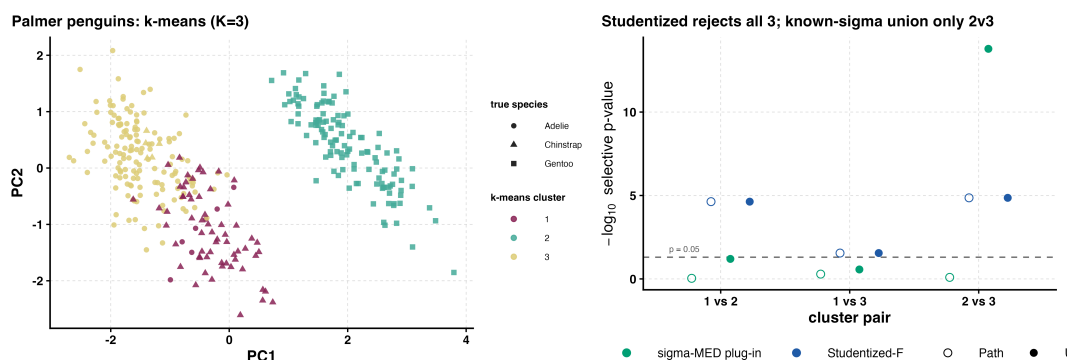


Figure 7: Palmer penguins, $K = 3$ (Table 4). Source: `penguins.png`.

- Lucy L. Gao, Jacob Bien, and Daniela Witten. Selective inference for hierarchical clustering. *Journal of the American Statistical Association*, 119(545):332–342, 2024. doi: 10.1080/01621459.2022.2116331. arXiv:2012.02936.
- Allison M. Horst, Alison P. Hill, and Kristen B. Gorman. palmerpenguins: Palmer archipelago (antarctica) penguin data, 2020. URL <https://allisonhorst.github.io/palmerpenguins/>. R package version 0.1.0.
- Jason D. Lee, Dennis L. Sun, Yuekai Sun, and Jonathan E. Taylor. Exact post-selection inference, with application to the lasso. *The Annals of Statistics*, 44(3):907–927, 2016. doi: 10.1214/15-AOS1371.
- Anna Neufeld, Ameer Dharamshi, Lucy L. Gao, and Daniela Witten. Data thinning for convolution-closed distributions. *Journal of Machine Learning Research*, 25(57):1–35, 2024. URL <https://jmlr.org/papers/v25/23-0446.html>. arXiv:2301.07276.
- Alex A. Pollen, Tomasz J. Nowakowski, Joe Shuga, Xiaohui Wang, Anne A. Leyrat, et al. Low-coverage single-cell mrna sequencing reveals cellular heterogeneity and activated signaling pathways in developing cerebral cortex. *Nature Biotechnology*, 32(10):1053–1058, 2014. doi: 10.1038/nbt.2967.
- Youngjoo Yun and Rina Foygel Barber. Selective inference for clustering with unknown variance. *Electronic Journal of Statistics*, 17(2):1923–1946, 2023. doi: 10.1214/23-EJS2143. arXiv:2301.12999.
- Youngjoo Yun and Yinqiu He. Selective inference for multiple pairs of clusters after k -means clustering. arXiv preprint, 2024. URL <https://arxiv.org/abs/2405.16379>. arXiv:2405.16379.

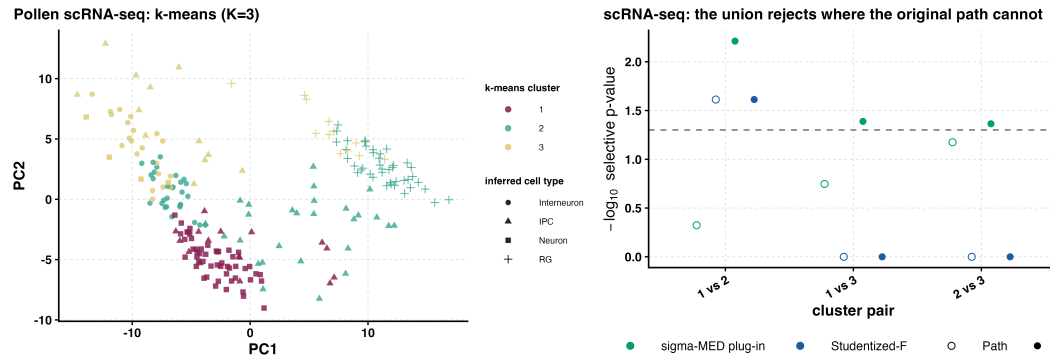


Figure 8: Pollen scRNA-seq, $K = 3$ (Table 5): the union rejects all three pairs, the path rejects none. Source: `scrna_k3.png`.